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313 HW4

7.D2)

0.6

K(34°C)

K(40°C)

Propylene: 1.00

1.13

Propane: 0.89

1.00

Isobutane: 0.42

0.46

Propylene > propane > Isobutane
LNK LK HK

$$\alpha = \frac{K_p}{K_r} = \frac{1.00}{0.46} = 2.17$$

$$\beta = \frac{x_{D,LK}}{x_{B,LK}} = \frac{0.05}{0.95} = 0.0526$$

$$M = \frac{x_{B,HK}}{x_{D,HK}} = \frac{0.9999}{0.0001} = 9999$$

$$N_{min} = \frac{\ln(\beta \cdot M)}{\ln \alpha}$$

$$N_{min} = \frac{\ln(0.0526 \times 9999)}{\ln(2.17)}$$

$N_{min} = 8.1$

8 theoretical stages

9 stages total

7.D7)

(a)

$$\text{CH}_4 = 7.75 \text{ kmol/h}$$

$$\text{C}_2\text{H}_6 = 15.5 \text{ kmol/h}$$

$$\text{n-C}_4 = 23.25 \text{ kmol/h}$$

$$\text{n-C}_5 = 34.1 \text{ kmol/h}$$

$$\text{n-C}_6 = 34.1 \text{ kmol/h}$$

$$\text{n-C}_7 = 40.3 \text{ kmol/h}$$

Distillate:

$$\text{Methane} \rightarrow 7.75$$

$$\text{Ethane} \rightarrow 15.5$$

$$\text{Butane} \rightarrow (0.99)(23.25) = 23.02$$

$$\text{Pentane} \rightarrow (0.07)(34.1) = 2.38$$

$$D = 46.85 \text{ kmol/h}$$

Bottoms:

$$\text{Butane} \rightarrow (0.01)(23.25) = 0.23$$

$$\text{Pentane} \rightarrow (0.983)(34.1) = 33.52$$

$$\text{Hexane} \rightarrow 34.1$$

$$\text{Heptane} \rightarrow 40.3$$

$$B = 108.15 \text{ kmol/h}$$

Distillate Composition:

$$\text{CH}_4 \rightarrow 0.165$$

$$\text{C}_2\text{H}_6 \rightarrow 0.331$$

$$\text{n-C}_4 \rightarrow 0.491$$

$$\text{n-C}_5 \rightarrow 0.012$$

Bottoms composition:

$$\text{n-C}_4 \rightarrow 0.002$$

$$\text{n-C}_5 \rightarrow 0.310$$

$$\text{n-C}_6 \rightarrow 0.315$$

$$\text{n-C}_7 \rightarrow 0.372$$

7.D7 cont)

(b) Component

K

Methane

13

Ethane

6

n-Butane

1.8

n-Pentane

1.0

n-Hexane

0.38

n-Heptane

0.15

$$\alpha_i = \frac{K_i}{K_5}$$

CH₄

13

C₂H₆

6

n-C₄

1.8

n-C₅

1.0

n-C₆

0.38

n-C₇

0.15

(c)

$$N_{min} = \frac{\ln \left(\frac{x_{D,LK}}{x_{B,LK}} \cdot \frac{x_{B,HK}}{x_{D,HK}} \right)}{\ln(\alpha_{LK/HK})}$$

$$= \frac{\ln(6536)}{\ln(1.8)}$$

$$N_{min} = 15$$

7.10) cont)

(d)

$$\sum \frac{z_i}{\alpha_i - \theta} = 1$$

$$\theta = 1.3$$

$$R_{min} = \sum \frac{x_{D,i} \alpha_i}{\alpha_i - \theta} - 1$$

$$R_{min} = 4.5$$

(e)

$$R = M R_{min}$$

M	R
1.04	4.68
1.10	4.95
2.0	9.0
M	N
1.04	~65
1.10	~48
2.0	~25

(f)

Likely not economical because,

- Very low distillate temperature
- Large number of stages
- Large reflux ratio
- Wide boiling point range mixture

7.D9)

(a)

$$\text{Benzene} = 78.0 \text{ kmol/h}$$

$$\text{Toluene} = 57.75 \text{ kmol/h}$$

$$\text{Xylene} = 14.25 \text{ kmol/h}$$

Distillate:

$$0.993 D = 78$$

$$D = 78.6 \text{ kmol/h}$$

$$0.007 (78.6) = 0.55$$

Bottoms:

$$B = F - D$$

$$B = 150 - 78.6 = 71.4$$

$$57.75 - 0.55 = 57.2$$

Distillate Composition:

$$\text{Benzene} \rightarrow 0.993$$

$$\text{Toluene} \rightarrow 0.007$$

$$\text{Xylene} \rightarrow \sim 0$$

Bottoms Composition:

$$\text{Benzene} \rightarrow 0$$

$$\text{Toluene} \rightarrow 0.801$$

$$\text{Xylene} \rightarrow 0.199$$

7.D9 cont)

(b) $L/v = 1$

Use McCabe - Thiele stepping

$N \approx 7$

(c)
$$N = \frac{\ln\left(\frac{x_{D,LK}}{x_{B,LK}} \cdot \frac{x_{B,HK}}{x_{D,HK}}\right)}{\ln(\alpha_{LK/HK})}$$

Using $N \approx 7$,

$x_{D,xy} \approx 2 \times 10^{-4}$

(d)

$$\sum \frac{z_i}{\alpha_i - \theta} = 1$$

$\theta = 1.45$

$$R_{min} = \sum \frac{x_{D,i} \alpha_i}{\alpha_i - \theta} - 1$$

$R_{min} = 1.6$

(e)

$R = 9$

$$\frac{R + R_{min}}{R + 1} = \frac{9 + 1.6}{10} = 0.74$$

$N = 8$

7.D13)

Benzene > Toluene > Cumene

LHK LK HK

Benzene $\rightarrow$ 25 mol/h

Toluene $\rightarrow$ 30 mol/h

Cumene $\rightarrow$ 45 mol/h

99% Toluene $\rightarrow$ distillate

= 29.7 mol/h

98% Cumene $\rightarrow$ Bottoms

= 44.1 mol/h

= 0.9 mol/h in Distillate

$$D = 25 + 29.7 + 0.9 = 55.6$$

$$B = F - D = 100 - 55.6 = 44.4$$

(a)

$$N_{min} = \frac{\ln\left(\frac{x_{D,LK}}{x_{B,LK}} \cdot \frac{x_{B,HK}}{x_{D,HK}}\right)}{\ln(\alpha_{LK/HK})}$$

$$N_{min} = \frac{\ln(4720)}{\ln(4.76)} = 5.4$$

$N=9$,

$$X = \frac{N - N_{min}}{N + 1} = \frac{9 - 5.4}{10} = 0.36$$

$$\sum \frac{z_i}{\alpha_i - \theta} = 1$$

$$\theta = 0.62$$

$$R_{min} = \sum \frac{x_{D,i} \alpha_i}{\alpha_i - \theta} - 1$$

$$R_{min} = 1.8$$

$$\frac{R - R_{min}}{R + 1} = 0.43$$

$$L/D = 3.3$$

7.D13 (cont)

(b) Using the same calculation process as part a, we get

$$L/D = 4.2$$

which makes sense, because lower volatility means harder separation. N_{min} increases, Gilliland shifts upward.

7.D14)

(a)

Ethanol $\rightarrow 30$ kmol/h

i-Propanol $\rightarrow 25$ kmol/h

n-Propanol $\rightarrow 35$ kmol/h

n-Butanol $\rightarrow 10$ kmol/h

Recoveries:

$$i\text{-Propanol} \rightarrow 0.986(25) = 24.65$$

$$n\text{-propanol} \rightarrow 0.992(35) = 34.72$$

$$= 0.28 \text{ distillate}$$

$$D = 30 + 24.65 + 0.28 = 54.93$$

$$B = 100 - 54.93 = 45.07$$

(b)

$$N_{min} = \frac{\ln \left(\frac{x_{D,LK}}{x_{B,LK}} \cdot \frac{x_{B,HK}}{x_{D,HK}} \right)}{\ln(\alpha_{LK/HK})}$$

$$N_{min} = 14.6$$

$$N_{F,min} = 9$$

7.1D/4 cent)
(C)

$$q=0$$

$$\sum_i \frac{z_i}{\alpha_i - \phi} = 1$$

$$\phi = 1.33$$

$$R_{min} = \sum_i \frac{x_{D,i} d_i}{\alpha_i - \phi} - 1$$

$$(4D)_{min} = 1.85$$

$$(d) \quad 4D = 1.10 (4D)_{min} = 2.04$$

$N = 24$ stages
$N_F = 15$